

Chapter 8

Hyperfine Structure for $L \neq 0$ and Isotope Shifts

The hyperfine interaction with electron orbital angular momentum is in §6.1.3 and the normal mass shift and the volume shift are covered in §6.2 of Foot [3].

8.1 Hyperfine interaction with electron orbital angular momentum

When we examined the effect of the magnetic dipole-dipole interaction between the electron spin in an s orbital and the nuclear spin, we found that the energy splitting this caused was due entirely to the part of the electron wavefunction that is contained within the nucleus. This so-called “Fermi contact interaction” was proportional to $|\psi(0)|^2$, and so we expect this to be zero for states with $L \neq 0$.

This does not, however, mean that non- s states will not show splittings due to magnetic interactions between the nucleus and the electron. In fact, unlike s states, the hyperfine interaction for $L \neq 0$ is strongly influenced by the electron’s spatial wavefunction outside of the nucleus, as we will show below.

To model the interaction, we will take the approach of considering the magnetic field produced by the electron at the position of the nucleus. This will potentially come from two sources: the magnetic field ($\mathbf{B}_S$) caused by the magnetic dipole of the electron’s spin, $\boldsymbol{\mu}_S$, and the magnetic field ($\mathbf{B}_O$)

caused by the current associated with the electron's orbit around the nucleus.

In both cases, we are going to be constructing position-representation operators whose expectation value in the atomic state of interest yields the magnetic field at the nucleus (which is the origin of our coordinate system). For $\mathbf{B}_S$, the operator will be the magnetic field at the origin in terms of the electron position operator $\hat{\mathbf{r}}_e$. For $\mathbf{B}_O$, we will first make an operator $\mathbf{j}_e$ (which will be written as a function of the electron position and momentum operators $\hat{\mathbf{r}}_e$ and $\hat{\mathbf{p}}_e$) whose expectation value will give the electric current at spatial position $\mathbf{r}$, then we will use the Biot-Savart law to integrate $\mathbf{j}_e(\mathbf{r}) \times \hat{\mathbf{e}}_r/r^2$ over all space $d^3\mathbf{r}$ to get an operator whose expectation value will give the magnetic field at the nucleus.

8.1.1 Electron spin contribution

For the electron spin contribution, we will start by writing the magnetic field at some position $\mathbf{R}$ due to a magnetic moment $\boldsymbol{\mu}_S$ at position $\hat{\mathbf{r}}_e$:

$$\mathbf{B}_S(\mathbf{R}, \hat{\mathbf{r}}_e) = -\frac{\mu_0}{4\pi} \left[\frac{\boldsymbol{\mu}_S - 3 \frac{(\mathbf{R}-\hat{\mathbf{r}}_e)}{|\mathbf{R}-\hat{\mathbf{r}}_e|} \left(\frac{(\mathbf{R}-\hat{\mathbf{r}}_e)}{|\mathbf{R}-\hat{\mathbf{r}}_e|} \cdot \boldsymbol{\mu}_S \right)}{|\mathbf{R} - \hat{\mathbf{r}}_e|^3} - \frac{8\pi}{3} \boldsymbol{\mu}_S \delta(\mathbf{R} - \hat{\mathbf{r}}_e) \right]. \quad (8.1)$$

Since we are interested in the field at the nucleus, we take $\mathbf{R} = 0$ to get

$$\mathbf{B}_S(0, \hat{\mathbf{r}}_e) = -\frac{\mu_0}{4\pi} \left[\frac{\boldsymbol{\mu}_S - 3 \hat{\mathbf{e}}_{r_e} (\hat{\mathbf{e}}_{r_e} \cdot \boldsymbol{\mu}_S)}{r_e^3} - \frac{8\pi}{3} \boldsymbol{\mu}_S \delta(\hat{\mathbf{r}}_e) \right]. \quad (8.2)$$

Since we are going to take the expectation value of this for an electron with $L \neq 0$, we see that the delta function is going to give $|\psi_{nLM}(0)|^2 = 0$, so we can drop it and we have

$$\hat{\mathbf{B}}_S(0, \mathbf{r}_e) = -\frac{\mu_0}{4\pi} \left[\frac{\boldsymbol{\mu}_S - 3 \hat{\mathbf{e}}_{r_e} (\hat{\mathbf{e}}_{r_e} \cdot \boldsymbol{\mu}_S)}{r_e^3} \right]. \quad (8.3)$$

8.1.2 Electron current contribution

For the electron orbit contribution to the magnetic field at the nucleus, we can start with the classical current density (electric current per unit area) at position $\mathbf{r}$, which is the charge density times the (possibly position-dependent) local velocity

$$\mathbf{j}_e(\mathbf{r}) = \rho_e(\mathbf{r}) \mathbf{v}_e \quad (8.4)$$

To write this as an operator whose expectation value will give the electrical current at spatial position $\mathbf{r}$, we can start with the charge density at position $\mathbf{r}$ due to an electron at position $\hat{\mathbf{r}}_e$ (which is an operator, not a unit vector),

$$\rho_e(\mathbf{r}, \hat{\mathbf{r}}_e) = -e \delta(\mathbf{r} - \hat{\mathbf{r}}_e). \quad (8.5)$$

Writing the velocity in terms of the translational momentum operator gives us¹

$$\mathbf{j}_e(\mathbf{r}, \hat{\mathbf{r}}_e, \hat{\mathbf{p}}_e) = -\frac{e}{m} \delta(\mathbf{r} - \hat{\mathbf{r}}_e) \hat{\mathbf{p}}_e. \quad (8.6)$$

The magnetic field operator, which should give the magnetic field at position $\mathbf{R}$ due to an electron at position $\hat{\mathbf{r}}_e$ can now be written using the Biot-Savart law,

$$\begin{aligned} \mathbf{B}_O(\mathbf{R}, \hat{\mathbf{r}}_e) &= \frac{\mu_0}{4\pi} \int d^3\mathbf{r} \frac{\mathbf{j}_e(\mathbf{r}, \hat{\mathbf{r}}_e, \hat{\mathbf{p}}_e) \times \frac{(\mathbf{R}-\mathbf{r})}{|\mathbf{R}-\mathbf{r}|}}{|\mathbf{R}-\mathbf{r}|^2} \\ &= -\frac{\mu_0}{4\pi} \frac{e}{m} \int d^3\mathbf{r} \frac{\delta(\mathbf{r} - \hat{\mathbf{r}}_e) \hat{\mathbf{p}}_e \times \frac{(\mathbf{R}-\mathbf{r})}{|\mathbf{R}-\mathbf{r}|}}{|\mathbf{R}-\mathbf{r}|^2} \\ &= -\frac{\mu_0}{4\pi} \frac{e}{m} \frac{\hat{\mathbf{p}}_e \times \frac{(\mathbf{R}-\hat{\mathbf{r}}_e)}{|\mathbf{R}-\hat{\mathbf{r}}_e|}}{|\mathbf{R}-\hat{\mathbf{r}}_e|^2} \end{aligned}$$

Last, since we're evaluating this at the nucleus, we set $\mathbf{R} = 0$ (and multiply the top and bottom by $r_e \equiv |\hat{\mathbf{r}}_e|$, which is an operator) to get

$$\begin{aligned} \mathbf{B}_O(0, \hat{\mathbf{r}}_e) &= -\frac{\mu_0}{4\pi} \frac{e}{m} \frac{\hat{\mathbf{p}}_e \times (-\hat{\mathbf{r}}_e)}{r_e^3} \\ &= -\frac{\hbar\mu_0 e}{4\pi m} \frac{\mathbf{L}}{r_e^3}. \end{aligned} \quad (8.7)$$

¹The careful reader is at this point cautioned that this expression for the quantum current density operator is not quite generally applicable. A more-thorough treatment [5] will yield a slightly more-symmetric expression for the operator whose expectation value gives the current density,

$$\mathbf{j}_e(\mathbf{r}, \hat{\mathbf{r}}_e, \hat{\mathbf{p}}_e) = -\frac{e}{2m} (\hat{\mathbf{p}}_e \delta(\mathbf{r} - \hat{\mathbf{r}}_e) + \delta(\mathbf{r} - \hat{\mathbf{r}}_e) \hat{\mathbf{p}}_e).$$

Since we do not need the expectation value of the current density directly (rather, we are interested in the magnetic field, which is proportional to $\mathbf{j} \times \hat{\mathbf{e}}_r$), it turns out that our “intuitive” version of the operator (8.6) will suffice for our purposes.

8.1.3 Interaction Hamiltonian

Combining the spin and orbit contributions allows us to write the total magnetic field at the nucleus in terms of the position ($\hat{\mathbf{r}}_e$) and angular momentum ($\mathbf{L}$) operators for the electron,

$$\begin{aligned}
\mathbf{B}(0, \hat{\mathbf{r}}_e) &= -\frac{\mu_0}{4\pi} \left[\frac{\boldsymbol{\mu}_S - 3\hat{\mathbf{r}}_e (\hat{\mathbf{r}}_e \cdot \boldsymbol{\mu}_S)}{r_e^3} + \frac{\hbar e}{m} \frac{\mathbf{L}}{r_e^3} \right] \\
&= -\frac{\mu_0}{4\pi} \left[-g_s \mu_B \frac{\mathbf{S} - 3\hat{\mathbf{r}}_e (\hat{\mathbf{r}}_e \cdot \mathbf{S})}{r_e^3} + \frac{\hbar e}{m} \frac{\mathbf{L}}{r_e^3} \right] \\
&= -2 \frac{\mu_0 \mu_B}{4\pi} \left(\frac{1}{r_e^3} \right) \left[\mathbf{L} - \frac{g_s}{2} (\mathbf{S} - 3\hat{\mathbf{r}}_e (\hat{\mathbf{r}}_e \cdot \mathbf{S})) \right] \\
&= -2 \frac{\mu_0 \mu_B}{4\pi} \left(\frac{1}{r_e^3} \right) \mathbf{K}
\end{aligned}$$

for vector operator

$$\mathbf{K} \equiv \mathbf{L} - \frac{g_s}{2} (\mathbf{S} - 3\hat{\mathbf{r}}_e (\hat{\mathbf{r}}_e \cdot \mathbf{S})). \quad (8.8)$$

The interaction Hamiltonian describing the energy of the nuclear magnetic dipole moment $\boldsymbol{\mu}_I$ in the magnetic field at the nucleus is

$$\begin{aligned}
H_{\text{hf}} &= -\boldsymbol{\mu}_I \cdot \mathbf{B}(0, \mathbf{r}_e) \\
&= 2g_I \mu_N \frac{\mu_0 \mu_B}{4\pi} \left(\frac{1}{r_e^3} \right) \mathbf{I} \cdot \mathbf{K}.
\end{aligned} \quad (8.9)$$

8.1.4 Expectation value

At this point, if we want to take the expectation value of (8.9), we are faced with needing to figure out what is happening with that operator $\mathbf{K}$. Unlike the case for s -states, where we were able to take a similar expectation value since we were approximately in an eigenstate of $\mathbf{L}^2$ and L_z due to L being zero (leaving $M_L = 0$ as the only possible choice), we don't have the information about the various $|L M_L\rangle$ and $|S M_S\rangle$ states making $|J M_J\rangle$ we would need to do this. We will hopefully cover that material soon, but for the moment, we're going to do our best to proceed without it. So we are going to once again make some plausibility arguments, and I am going to pull a result from the more complicated mathematics without justifying it to get us to the point

where we can evaluate the expectation value of this hyperfine Hamiltonian for non- s states.

The plausibility argument will start with the question: in which direction can this field, $\langle \mathbf{B}(0, \hat{\mathbf{r}}_e) \rangle$, point? Foregoing the hyperfine splitting itself for a moment and considering that the atom is in a state $|J M_J\rangle$, what do we know about it?

The answer is that we know $\langle \mathbf{J}^2 \rangle = J(J+1)$ and $\langle J_z \rangle = M_J$. If we try to evaluate the expectation value of any component of $\mathbf{J}$ that is perpendicular to $\hat{\mathbf{e}}_z$, we will get zero. The angular momentum cannot point in any direction, on average, other than $\pm \hat{\mathbf{e}}_z$ for an atom in a state $|J M_J\rangle$.

And it turns out that this is true for *any* vector operator $\mathbf{V}$, whether we imagine $\mathbf{V} = \mathbf{S}$, or $\mathbf{V} = \mathbf{L}$, or $\mathbf{p}$ or whatever you like. After all, there is nothing special about $\hat{\mathbf{e}}_x$ or $\hat{\mathbf{e}}_y$, and the Hamiltonian itself doesn't identify any preferred direction in space. We chose to *define* $\hat{\mathbf{e}}_z$ so that the states were in a J_z eigenstate, but we also showed that (since orthogonal components of $\mathbf{J}$ don't commute) this cannot also be an eigenstate of J_x or J_y .

We come, then, to the following conclusion:

- *The expectation value of any vector operator in a total angular momentum eigenstate has to point the same direction as the expectation value of the total angular momentum.*

This is called the *Wigner-Eckart projection theorem*, and its proof is reasonably accessible, but beyond the scope of this course.

We therefore conclude that $\langle \mathbf{B}(0, \hat{\mathbf{r}}_e) \rangle$ (and, by extension, $\langle \mathbf{K} \rangle$) can be written as a scalar constant (call it β) times $\langle \mathbf{J} \rangle$. Any expression, then, proportional to any component of $\mathbf{K}$ that is perpendicular to $\mathbf{J}$ must also yield 0 for its expectation value.

This argument has given us some confidence that we can write our Hamiltonian in the form

$$H_{\text{hf}} = -\boldsymbol{\mu}_I \cdot \mathbf{B}(0, \hat{\mathbf{r}}_e) \quad (8.10)$$

$$= 2g_I\mu_N \frac{\mu_0\mu_B}{4\pi} \left(\frac{1}{r_e^3} \right) \mathbf{I} \cdot (\beta\mathbf{J}) \quad (8.11)$$

for some scalar constant β . Taking $\beta \approx 1$ as reasonable (after all, the form of $\mathbf{K}$ is somewhat suggestive of $\mathbf{K} \approx \mathbf{J}$, and it seems unlikely that $\mathbf{K} = 1000\mathbf{J}$, a claim that you will explore on your problem set this week), we get the

expectation value

$$\begin{aligned}
 E_{\text{hf}} &\approx \langle H_{\text{hf}} \rangle \\
 &\approx 2g_I \mu_N \frac{\mu_0 \mu_B}{4\pi} \left\langle \frac{1}{r_e^3} \right\rangle \langle \mathbf{I} \cdot \mathbf{J} \rangle \\
 &= 2g_I \mu_N \frac{\mu_0 \mu_B}{4\pi} \frac{1}{a_0^3 L(L + \frac{1}{2})(L + 1)} \frac{Z^3}{n^3} \langle \mathbf{I} \cdot \mathbf{J} \rangle. \quad (8.12)
 \end{aligned}$$

So here, again, we find that we will get an interval rule in the total angular momentum quantum number F since $\mathbf{F} = \mathbf{J} + \mathbf{I}$ and the energy shift goes like $\mathbf{I} \cdot \mathbf{J}$. Much like the Fermi contact interaction, this Hamiltonian is also often written in the spectroscopic form

$$H_{\text{hf}} = A_{\text{hf}} \mathbf{I} \cdot \mathbf{J}. \quad (8.13)$$

8.2 Isotope Shifts

The Bohr model provided us with a way to find the difference between the energies of states in hydrogen and deuterium by using the reduced mass in the Bohr formula. This was our first brush with what is called an *isotope shift*. Isotope shifts are changes in atomic energies that occur when comparing isotopes, which means species with the same number of protons (Z) but a different number of neutrons (N). However, isotope shifts do *not* include hyperfine effects. As you will show on this week's problem set, hyperfine structure at our level of analysis does not change the average energy of an electronic state once one averages over all of the possible $|F M_F\rangle$ states, but isotope shifts can. Another way to put it is that hyperfine interactions cause *splittings* of n, J manifolds into states with different energies whereas isotope shifts can only *shift* the average energy of each n, J manifold.

8.2.1 Normal Mass shift

As an example of an isotope shift, the *normal mass shift* is the isotope shift we got from the difference in reduced mass of different isotopes. Since we already covered this in §2.3, there is no need to rehash it here.

8.2.2 Volume Shift

Another source of an isotope shift, the *volume shift*, comes from the fact that the electric potential inside the nucleus due to the electron depends upon r , and the nucleus has finite size. In this section, we will calculate the volume shift for a hydrogenic electron in an s state.

For an $L = 0$ state, the electron charge distribution is spherically symmetric, and we can calculate the electric field at radial distance r due to the electron using Gauss's law,

$$\mathbf{E}_e(\mathbf{r}) = \frac{q_{\text{enc}}(r)}{4\pi\epsilon_0 r^2} \hat{\mathbf{e}}_r. \quad (8.14)$$

where $q_{\text{enc}}(r)$ is the electron charge enclosed in a sphere of radius r .

Since there will be no contribution to the field in the nucleus from the spherically-symmetric charge distribution outside of the nucleus, we will again utilize the approximation that the electron density is constant over the size of the nucleus, which gives a charge density of

$$\rho_e(\mathbf{r}) = -e |\psi_{n00}(\mathbf{r})|^2 \approx -e |\psi_{n00}(0)|^2. \quad (8.15)$$

The charge enclosed in a sphere of radius r , then, is

$$q_{\text{enc}}(r) = \frac{4}{3}\pi r^3 \rho_e \quad (8.16)$$

and the electric field is

$$\mathbf{E}_e(\mathbf{r}) = \frac{\rho_e}{3\epsilon_0} \mathbf{r}. \quad (8.17)$$

An electric potential that yields this field is given by

$$\Phi_e(r) = -\frac{\rho_e r^2}{6\epsilon_0} \quad (8.18)$$

$$= \frac{e r^2}{6\epsilon_0} |\psi_{n00}(0)|^2 \quad (8.19)$$

To get the interaction energy with the nucleus, we can integrate the product of the electric potential from the electron and the nuclear charge density, which is due to the protons, $\rho_p(r)$,

$$E_{\text{VS}} = \int d^3\mathbf{r} \Phi_e(r) \rho_p(r) \quad (8.20)$$

$$= \frac{e}{6\epsilon_0} |\psi_{n00}(0)|^2 \int d^3\mathbf{r} r^2 \rho_p(r) \quad (8.21)$$

The proton charge distribution can be written as

$$\rho_p(\mathbf{r}) = Ze |\psi_{\text{protons}}(\mathbf{r})|^2 \quad (8.22)$$

where $\psi_{\text{protons}}(\mathbf{r})$ is the (likely unknown) wavefunction of the protons. This allows us to write the integral as an expectation value of the nuclear charge distribution,

$$\int d^3\mathbf{r} r^2 \rho_p(r) = Ze \langle r_p^2 \rangle \quad (8.23)$$

and we have

$$E_{\text{VS}} = Z \frac{e^2}{6\epsilon_0} |\psi_{n00}(0)|^2 \langle r_p^2 \rangle \quad (8.24)$$

$$= \frac{e^2}{\epsilon_0} \frac{Z^4}{6\pi a_0^3 n^3} \langle r_p^2 \rangle. \quad (8.25)$$

Using

$$R_\infty = \frac{e^2}{8\pi\epsilon_0 a_0 hc}, \quad (8.26)$$

we can write this as

$$E_{\text{VS}} = \frac{4}{3} \frac{\langle r_p^2 \rangle}{a_0^2} hc R_\infty \frac{Z^4}{n^3}. \quad (8.27)$$

So by measuring the isotope shift between two atoms with different numbers of neutrons, the change in rms size of the proton distribution can be inferred.